2 Sequencing Puzzle — Morning Routine

AGES 8-12

MASCOT GOES HERE

KID INSTRUCTIONS

Last time, you wrote an algorithm for making a sandwich. But what if you spread the peanut butter **BEFORE** you got the bread out? Algorithms only work when the steps are in the right order — let's see what happens when they're not!

Sequence

Sequence — The exact order that steps happen in, from first to last.
Example: Your morning is a sequence — you wake up, then brush your teeth, then eat breakfast. Switch the order and your day gets weird fast!

Basic Version

Fix the Mixed-Up Morning — These morning steps got all jumbled up! Number them 1 to 8 in the correct order so the morning actually works.

- ☐ Put on shoes
- ☐ Wake up
- ☐ Brush teeth
- ☐ Eat breakfast
- ☐ Get dressed
- ☐ Pack school bag
- ☐ Wash face
- ☐ Go to school

Challenge Version

Spot the Broken Sequence — Each algorithm below has **ONE** step in the wrong place. Find it, circle it, and write where it should go.

Algorithm A: Making Cereal

1. Get a bowl from the cupboard
2. Pour the cereal into the bowl
3. Eat the cereal
4. Pour milk on the cereal
5. Get a spoon

Which step is in the wrong place? Where should it go?

Algorithm B: Getting Ready for Soccer Practice

1. Put on your soccer cleats
2. Put on your soccer socks
3. Put on your jersey and shorts
4. Grab your water bottle
5. Head to the field

Which step is in the wrong place? Where should it go?

Reflection Box

What happens if you put on your shoes **BEFORE** your socks? Why does order matter for algorithms?

Self-Check

- [ ] I numbered all 8 morning steps from 1 to 8
- [ ] I found the wrong step in both Challenge algorithms
- [ ] I explained where the wrong step should go
- [ ] I can explain why order matters in my own words

Bonus: Cut & Sort — Brushing Your Teeth

Cut out these 6 steps and glue them in the correct order on a separate piece of paper.

CUT ALONG THE DASHED LINES

Spit out the toothpaste and rinse your mouth	Pick up your toothbrush	Brush your teeth for 2 minutes
Put toothpaste on the toothbrush	Wet the toothbrush under the tap	Rinse the toothbrush and put it back

TEACHER / PARENT NOTE

This activity introduces sequencing — the CS principle that the same instructions in a different order produce different results (or no result at all). In real code, sequence bugs are one of the most common errors beginners hit: variables used before they're defined, files saved before they're written, etc. Discussion prompt: Ask your child to think of a time when doing things in the wrong order caused a problem in real life — putting on a coat before a sweater, sending a text before reading it, etc. The connection between everyday sequence mistakes and code bugs makes the concept stick.